

Coil Design Analysis

This section analyses various coil structures and their properties, with respect to frequency of operation and distance between the transmitting and receiving coil. The circuit is shown in Figure 1. The same LC compensation circuit is used for all coil designs. Magnetic Flux is the measure of magnetic field which passes through a given area. Some key terms related to WPT are given below. Coupling coefficient is the fraction of magnetic flux produced by the current in one coil that links the other coil. Efficiency is the percentage of power received at Rx with respect to power given at Tx. Voltage Gain is the ratio of the input voltage to output voltage. Resonance is the phenomenon when reactance of capacitor and inductor are equal and opposite. Maximum power is transferred when resonance occurs. Resonant frequency is the frequency at which resonance occurs. In the following study, a comparison is made between different coil structures of similar dimension. Similar LC compensation circuit is used to compare efficiency and voltage gain as a function of frequency and coupling coefficient, self-inductance and mutual inductance as a function of distance. Magnetic field magnitude and vector have been plotted in plane of Rx. All simulations have been done in Ansys Maxwell and circuits have been simulated in Simplorer, with Maxwell model integrated.

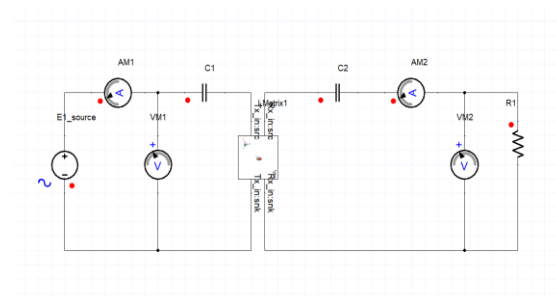


Fig.1 MODEL CIRCUIT DIAGRAM

A. HELICAL COIL:

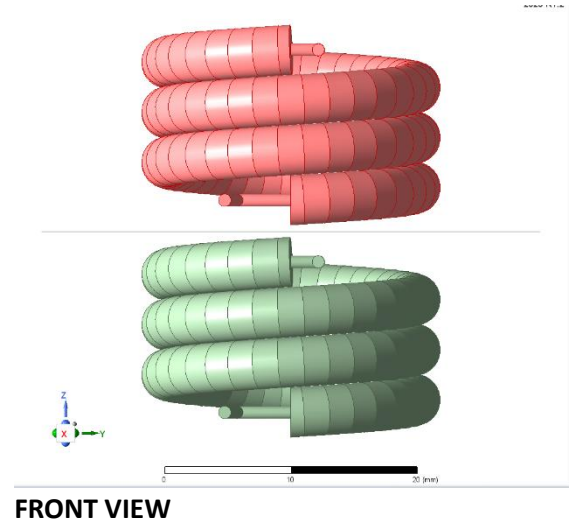
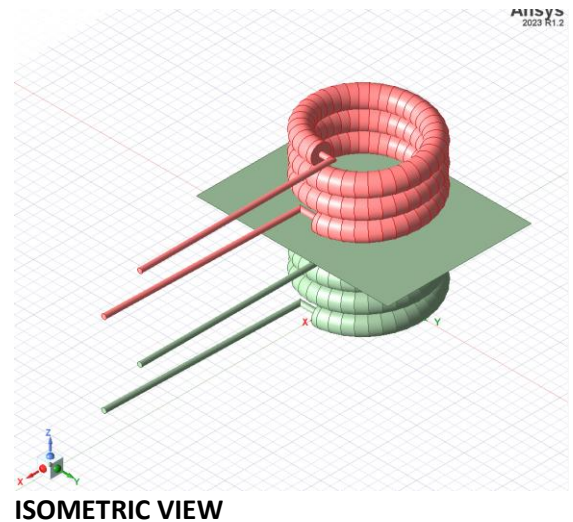


Fig. model for helix coil

1. Parameter

The frequency of operation is 6.78Mhz. Some parameters of coil are as tabulated in the given table

Properties: WPT TEST SYS for helix coil - Maxwell3DDesign1 - Modeler					
Command					
Name	Value	Unit	Evaluate	Description	
Command	CreateUserDefinedPart				
Coordinate System	Global				
Name	SegmentedHelix/PolygonHelix.dtl				
Location	syslib				
Version	1.0				
PolygonSegments	0		0	Number of cross-section	
PolygonRadius	2	mm	2mm	Outer radius of cross-section	
StartHelixRadius	10	mm	10mm	Start radius from polygon	
RadiusChange	0	mm	0mm	Radius change per turn	
Pitch	4.02	mm	4.02mm	Helix pitch	
Turns	3		3	Number of turns	
SegmentsPerTurn	36		36	Number of segments per turn	

2. INDUCTANCE VALUES :

The inductance values are tabulated in Table 16. We can observe that mutual inductance decreases with distance

L Table 1

	dist [mm]	L(Rx_in,Rx_in) [nH] Setup1 : LastAdaptive	L(Tx_in,Rx_in) [nH] Setup1 : LastAdaptive	L(Rx_in,Tx_in) [nH] Setup1 : LastAdaptive	L(Tx_in,Tx_in) [nH] Setup1 : LastAdaptive
1	17.000000	212.347216	19.706105	19.706105	206.523654
2	20.000000	212.327491	12.585317	12.585317	207.032590
3	25.000000	210.517433	6.108703	6.108703	207.497361
4	30.000000	208.974347	2.898733	2.898733	207.008239
5	35.000000	189.021087	1.015509	1.015509	207.448632

Fig. 3: graph of efficiency vs freq

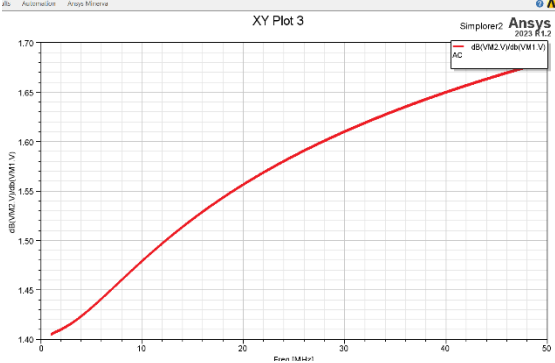


fig.4

3 **RESULTS:** The coupling coefficient decreases with distance as shown in Figure 2.

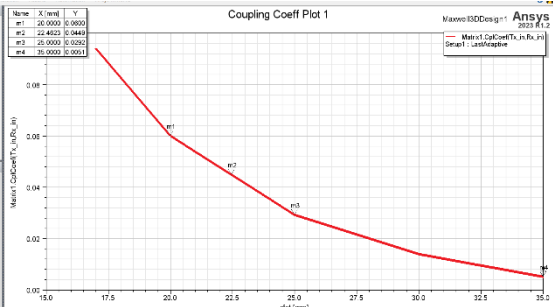


Fig. 2: Coupling Coefficient vs distance graph

The efficiency and voltage gain plots are shown in Fig 3 and Fig 4. The efficiency and voltage gain are broad

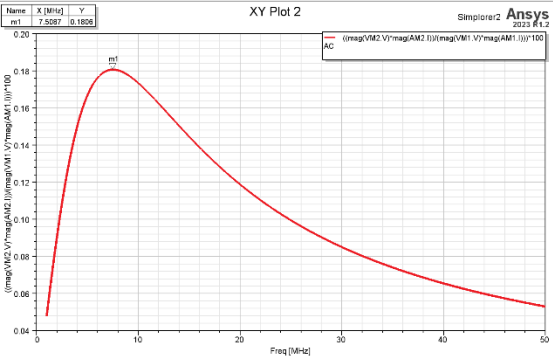


fig. 3

3. **FIELDS:** The magnetic field variation is shown in Figure 5 and Figure 6

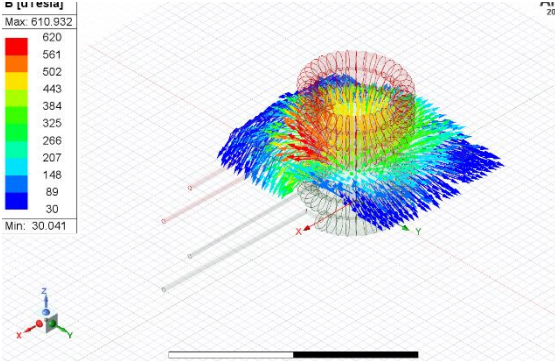


fig.5

Fig. 20: Magnetic field vector

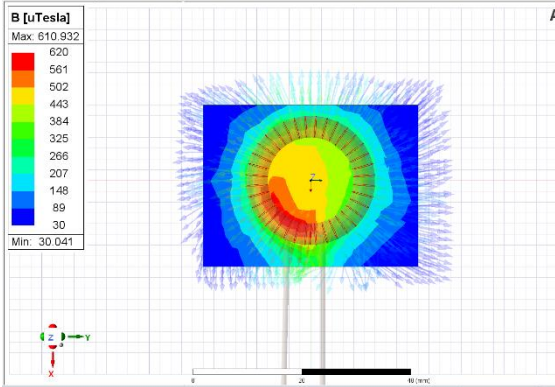


Fig. 7: Magnitude of Magnetic field

B. Rectangular spiral coil

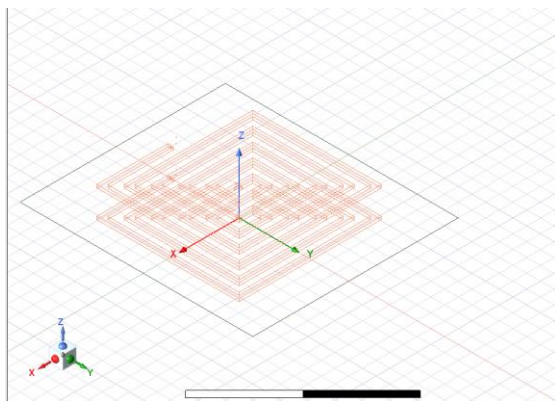


fig.7 isometric view

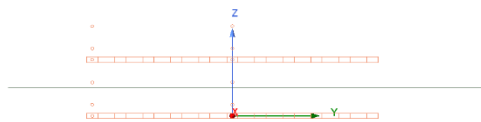


fig.8 Front View

1.PARAMETERS: The frequency of operation is 6.78Mhz. Some parameters of coil are as tabulated in the given table

Properties: wpt system: SPIRAL COIL - wpt spiral coil - Moduler				
Command				
Name	Value	Unit	Evaluated	Description
Coordinate System	Global/Local/Default			
Name	Example/Hexagon			
Location	xy/z			
Version	2.0			
Xpos	0	mm	0mm	X Position of start point
Ypos	0	mm	0mm	Y Position of start point
Dist	5	mm	5mm	Distance between turns
Turns	5			Number of turns
Width	2	mm	2mm	Width of the spiral
Thickness	1	mm	1mm	Thickness/height of the spiral

2.INDUCTANCE VALUES: The inductance values are tabulated in Fig 24. We can observe

L Table 1					
wpt spiral coil					
dist [mm]	Ltbl_inch_in [uH]	Ltbl_inch_in [nH]	Ltbl_inch_in [pH]	Ltbl_inch_in [fH]	Ltbl_inch_in [aH]
1	9.050000	1.046966	529.454185	529.454180	1.091160
2	10.000000	1.058873	351.549885	351.549885	1.074287
3	15.000000	1.055347	244.227648	244.227648	1.078900
4	25.000000	1.055486	128.506595	128.506595	1.090680
5	35.000000	1.051208	74.580488	74.580488	1.093128
6	45.000000	1.056247	46.885510	46.885510	1.117521
7	50.000000	1.059495	37.709695	37.709695	1.116983

Fig. 24: Inductance values

3.RESULTS: The coupling coefficient decreases with distance as shown in Figure 25.

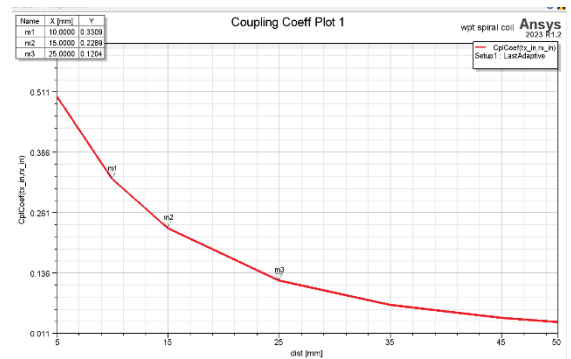


Fig. 25: Coupling Coefficient vs distance graph

The efficiency plot is shown in Figure 26. We can observe a sharp rise in the efficiency near the resonant frequency.

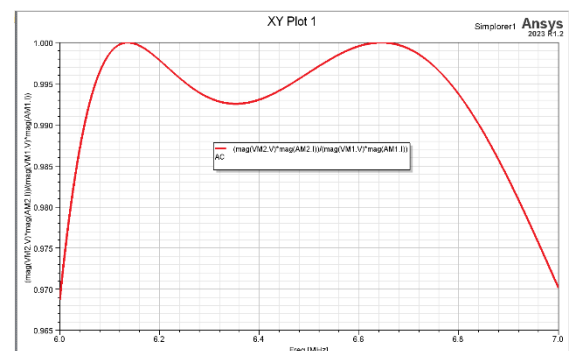


Fig. 26: Efficiency vs Fre

4) FIELDS: The magnetic field variation is shown in Figure 27 and Figure 28. The linkage is concentrated towards the center

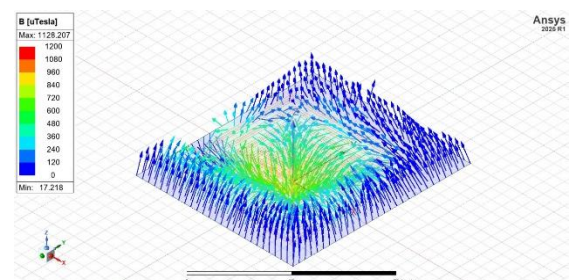


Fig.27: Magnetic Field Vector

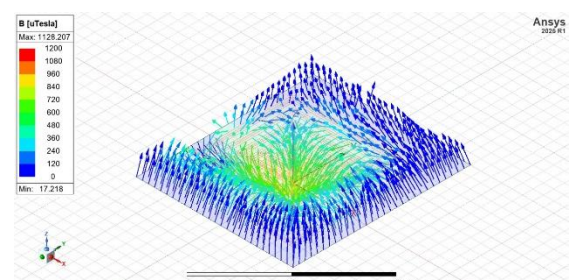


Fig.28: Magnetic Field